

Business Finance Management System

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Project Management 3

Project Portfolio

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Project Charter

Business Finance Management Project

Date 10 October 2025

Scope Overview:

The Business Finance Management System (BFMS) will be scalable and secure and will support firms with budgeting, income, expenses, payroll, and financial reporting. It will provide user authentication with role-based access in such a way that admins and staff will be able to view financial transactions, categorize expenses, and generate invoices. The system will accommodate budgeting, payroll, and auto-calculation of taxes in real-time with real-time financial information dashboards. The BFMS will be programmed using HTML, JavaScript, and CSS for the front-end, Node.js for the back end, and SQL for the database.

Vision Statement:

Business Finance Management System (BFMS) is an extensive web-based application by which organizations can monitor their finances, such as monitoring income, expense management, budgeting, payroll, and financial reporting. The target audience of the product is small and medium-sized enterprises, start-ups, and solo business owners who need an efficient and automated system to monitor their finances. These types of organizations tend to experience manual financial monitoring, budgeting errors, and lengthy payroll operations that require BFMS as a necessity for increased efficiency and accuracy.

With BFMS, businesses will save time, reduce financial errors, receive real-time insights, and stay tax compliant. The simple-to-use dashboard, automated capabilities, and secure payment processing of the system will provide businesses seamless financial management so that they can focus on growth rather than paperwork. BFMS stands out from other solutions as it has a simple-to-use interface, is affordable, provides real-time analytics, and integration so that even non-accounting experts can handle their business finances without hassle.

Business Case:

Introduction

Business financial management is a crucial yet challenging function, especially for small and medium-sized enterprises (SMEs) with no dedicated finance departments. Most firms utilize outdated manual processes or patchwork software, thus inefficiency, mistakes, and

compliance issues. Business Finance Management System (BFMS) seeks to streamline financial management through a single system to manage income, expenses, salaries, and financial reporting. Through the provision of automation, security, and real-time visibility, BFMS enables businesses to become more efficient and make better financial decisions. The project aligns with the growing need for digitalization of financial management so that SMEs can achieve financial stability and transparency.

Business Objective

The overall goal of Business Finance Management Systems (BFMS) is to simplify financial management by having a user-friendly and automated system for transaction recording, reporting, and payroll and invoice management. Through the system, small and medium sized enterprises will be able to reduce financial errors, save time, stay compliant, and make informed decisions with timely financial information. Designed to be scalable, the system will support business growth by accommodating increased transaction volumes and additional users as Small and medium-sized enterprises (SMEs) expand. Ultimately, the BFMS aligns with the broader strategic goal of helping SMEs embrace digital transformation, achieve financial stability, and remain competitive in a dynamic business environment.

Current Situation and Problem/Opportunity Statement

Small and medium-sized enterprises (SMEs) continue to face significant challenges in managing their finances due to heavy reliance on manual processes such as spreadsheets and paper-based tracking. These outdated methods are time-consuming, error-prone, and disruptive, often leading to inaccurate reporting, missed deadlines, and non-compliance with regulatory requirements. The absence of integrated financial systems also means that managers lack real-time visibility into income, expenses, and cash flow, making it difficult to plan budgets effectively or respond quickly to financial risks. As a result, many SMEs experience inefficiencies that not only increase operational costs but also divert valuable resources away from strategic growth activities. The opportunity lies in adopting an automated, user-friendly, and cost-effective platform such as the Business Finance Management System (BFMS), which consolidates budgeting, payroll, invoicing, and reporting into a single solution. By addressing inefficiency, improving accuracy, and ensuring compliance, the BFMS enables SMEs to operate more efficiently, gain timely insights, and shift focus from administrative burdens to growth and long-term competitiveness.

Critical Assumptions and Constraints

It is assumed that small and medium-sized enterprises (SMEs) have a strong demand for affordable, low-complexity financial management solutions that reduce reliance on manual processes. It is further assumed that organizations will adopt the system if it offers simplicity, automation, and tangible efficiency gains without requiring extensive training or technical expertise. Another assumption is that SMEs are increasingly open to cloud-based platforms if data protection and compliance are guaranteed, as this offers flexibility and cost savings. The project will, however, operate within several constraints. Data security and regulatory compliance are critical limitations that must be addressed to build trust and ensure legal adherence. The system must also be scalable, capable of supporting SMEs as they expand in size and transaction volume. Finally, the project faces budgetary constraints for both initial development and ongoing maintenance, requiring cost-effective design choices and prioritization of essential features.

Analysis of Options and Recommendation

The options for the project are to carry on with manual financial management, which is prone to high likelihood of errors, inefficiency, and non-compliance; make use of third-party available solutions, which might be non-cost-effective and non-customizable to small and medium-sized enterprises (SMEs) or create an in-house tailored Business Finance Management System (BFMS), which has tailor-made features, more control, and affordability. The path ahead is to proceed and execute Option 3 and build the BFMS using HTML, JavaScript, CSS, Node.js, and SQL, and make it efficient, affordable, and scalable for SMEs.

Preliminary Project Requirement

The Business Finance Management System (BFMS) will include a set of core functional and non-functional requirements to ensure usability, security, and compliance. At the foundation, the system will provide secure user authentication with role-based access control, ensuring that employees, managers, and administrators have appropriate permissions. Core financial features will include income and expenditure management, with categorized transactions to support accurate budgeting and forecasting. The system will also incorporate payroll management with automated calculations, reducing errors and administrative workload, as well as billing and invoicing tools with payment tracking to streamline cash flow management. To support decision-making, the BFMS will provide real-time financial reporting and interactive dashboards, giving businesses clear insights into performance, expenses, and

profitability. In addition, regulatory compliance checks and built-in data encryption will be integrated to safeguard sensitive financial information and ensure adherence to relevant legal standards. Collectively, these requirements will establish a secure, user-friendly, and scalable platform tailored to the needs of SMEs.

Budget Estimate and Financial Analysis

Estimated Budget:

- Development costs: R258 000.00
- Hosting and maintenance: R3 000.00 annually
- Marketing and onboarding support: R10 000.00
- Equipment: R40 000 .00

Financial Benefits:

The benefits of the Business Finance Management System include cost savings by reducing manual processes and minimizing human errors, improved financial accuracy and compliance with regulations, and faster decision-making through automated financial reports that provide real-time insights.

Potential Risks

The Business Finance Management System (BFMS) faces several potential risks that could impact its successful implementation and operation. Security threats remain a primary concern, as financial systems handle sensitive information and are vulnerable to cyber-attacks, data breaches, and unauthorized access. User adoption challenges are another risk, particularly for companies transitioning from manual processes or other software, as staff may resist change or lack the technical skills to use the system effectively. Scalability issues may arise as organizations grow, requiring the system to handle more users, higher transaction volumes, and more complex financial operations. Regulatory compliance risks are also significant, as changes in tax laws or financial reporting standards could require system updates. Furthermore, budget overruns may occur if scope creep introduces additional features beyond the original plan. Finally, there is a dependency on key personnel, such as developers or system administrators, whose absence or turnover could disrupt development, maintenance, or support.

Background:

Selection and Prioritization Matrix

Project/Selection Criteria	Impact		Feasibility		Efficiency		Risk Level		Total
	10		5		10		5		
Business Finance Management System	9		4		9		5		225
	90		20		90		25		
Employee Management System	8		5		8		5		210
	80		25		80		25		
University Management System	8		5		8		3		200
	80		25		80		15		
Hospital Management System	7		5		8		4		195
	70		25		80		20		

Figure 1 Selection and Prioritization Matrix

1. Business Impact – This quality measures the amount of value created by the project for the organization or users. High business impact projects increase efficiency, generate higher revenue, or improve customer satisfaction. Since creating effective and productive systems is the goal, this quality gets priority over others.
2. Feasibility – A project must be possible and achievable within the time, resources, and technical ability available. If a project is of excessive complexity or requires unavailability resources, it may not be viable. Feasibility is necessary to complete the project successfully. Weight is less as feasibility is required but not the most critical one.
3. Cost Efficiency – It will have to be checked whether the project is strongly return on investment. If the project is expensive but will not reap enough benefits, then it may not be worth it. Cost efficiency will ensure that there will be no waste of resources. Projects with high value for low cost are highly rated. This weight is high to make sure cost-effective solutions have more priority.
4. Risk Level – Every project has some risks associated with it, such as security risks, compliance risks, or technology failures. Awareness of these risks helps us plan to avoid them and ensures the success of the project. An extremely risky project would require additional protective steps, affecting its feasibility. More risky projects have lower scores.

NPV Table

Business Finance Management System

	Year 0	Year 1	Year 2	Year 3	Year 4
Anticipated Savings	R0	R80 000	R90 000	R110 000	R130 000
Project Cost	R258 000	R0	R0	R0	R0
Future Value	R258 000	R80 000	R90 000	R110 000	R130 000
Present Value	R258 000	R72 727	R74 380	R82 645	R88 795
Net Present Value	R60 547				

Figure 2 Business Management System NPV Table

Employee Management System

	Year 0	Year 1	Year 2	Year 3	Year 4
Anticipated Savings	R0	R90 000	R90 000	R90 000	R90 000
Project Cost	R250 000	R0	R0	R0	R0
Future Value	R250 000	R90 000	R90 000	R90 000	R90 000
Present Value	R250 000	R81 818	R74 380	R67 619	R61 471
Net Present Value	R35 288				

Figure 3 Employee Management System NPV Table

University Management System

	Year 0	Year 1	Year 2	Year 3	Year 4
Anticipated Savings	R0	R50 000	R50 000	R50 000	R50 000
Project Cost	R120 000	R0	R0	R0	R0
Future Value	R120 000	R50 000	R50 000	R50 000	R50 000
Present Value	R120 000	R45 455	R41 322	R37 566	R34 151
Net Present Value	R38 494				

Figure 4 University Management System NPV Table

Hospital Management System

	Year 0	Year 1	Year 2	Year 3	Year 4
Anticipated Savings	R0	R35 000	R35 000	R35 000	R35 000
Project Cost	R80 000	R0	R0	R0	R0
Future Value	R80 000	R35 000	R35 000	R35 000	R35 000

Present Value	R80 000	R31 818	R28 926	R26 296	R23 905
Net Present Value	R30 945				

Figure 5 Hospital Management System NPV Table

Scope Overview:

The Business Finance Management System (BFMS) is a comprehensive, scalable, and secure web-based platform designed to assist small and medium-sized enterprises (SMEs) in effectively managing their financial operations. The system aims to automate and streamline critical financial processes, including budgeting, income and expense tracking, payroll management, tax calculation, and financial reporting. By centralising these activities, the BFMS will significantly reduce manual effort, improve data accuracy, and enhance overall operational efficiency.

The system will incorporate role-based user authentication, ensuring that each user has appropriate access based on their role within the organisation. Administrators will have full access to manage users, approve transactions, and generate reports, while staff members will be able to record financial transactions, categorise expenses, and issue invoices. This structure promotes data security, transparency, and accountability across all levels of the business.

A key feature of the BFMS will be its automated payroll and tax calculation modules, designed to help businesses maintain compliance with financial regulations while saving time on routine administrative tasks. The system will also include interactive dashboards that provide real-time insights into income, expenditure, and cash flow trends, supporting informed decision-making and proactive financial planning.

Technically, the system will be built using HTML, CSS, and JavaScript for the front-end interface to ensure usability and responsiveness. The back end will be powered by Node.js for speed and scalability, while a SQL database will store all financial data securely. Built-in features such as audit trails, data encryption, and cloud-based hosting will ensure that all transactions are secure, traceable, and accessible from anywhere.

Overall, the BFMS will enable SMEs to modernise their financial management, reduce human error, ensure compliance, and focus more on business growth rather than manual financial administration.

Boundary Definition

The Business Finance Management System (BFMS) project will focus exclusively on developing and deploying a web-based financial management platform that supports

budgeting, expense tracking, payroll, invoicing, and reporting. The project scope is limited to system design, development, testing, deployment, and user training.

In Scope:

- Development of budgeting, payroll, invoicing, and reporting modules.
- Implementation of role-based access control and secure authentication.
- Creation of real-time dashboards and automated tax calculations.
- User documentation, testing, and initial deployment support.

Out of Scope:

- Hardware procurement and setup for client users.
- Long-term maintenance and technical support beyond project closure.
- Integration with third-party systems not specified in requirements.
- Development of native mobile applications (web version only for this release).

Project & Milestone Schedule

Current State:

Currently, small and medium-sized enterprises (SMEs) rely heavily on manual processes, spreadsheets, and fragmented software to manage finances. Transaction processing, payroll, budgeting, and invoicing are time-consuming and prone to errors, leading to inaccurate financial reporting and compliance risks. Managers lack real-time visibility into cash flow, expenses, and revenue, which makes informed decision-making difficult and slows down strategic planning.

Future State:

Upon completion of all milestones, SMEs will operate with a fully integrated Business Finance Management System that automates transaction processing, payroll, and invoicing; provides real time financial reporting and ensures compliance with financial regulations. Manual errors and redundant processes will be significantly reduced, and financial data will be readily accessible for informed decision-making.

Goal:

- Reduce manual processing time by 50%.
- Lower financial reporting errors by 70%.
- Achieve positive cash payback within 3 years.
- Secure at least 80% user adoption within 6 months of deployment.

Work Break Down Structure:

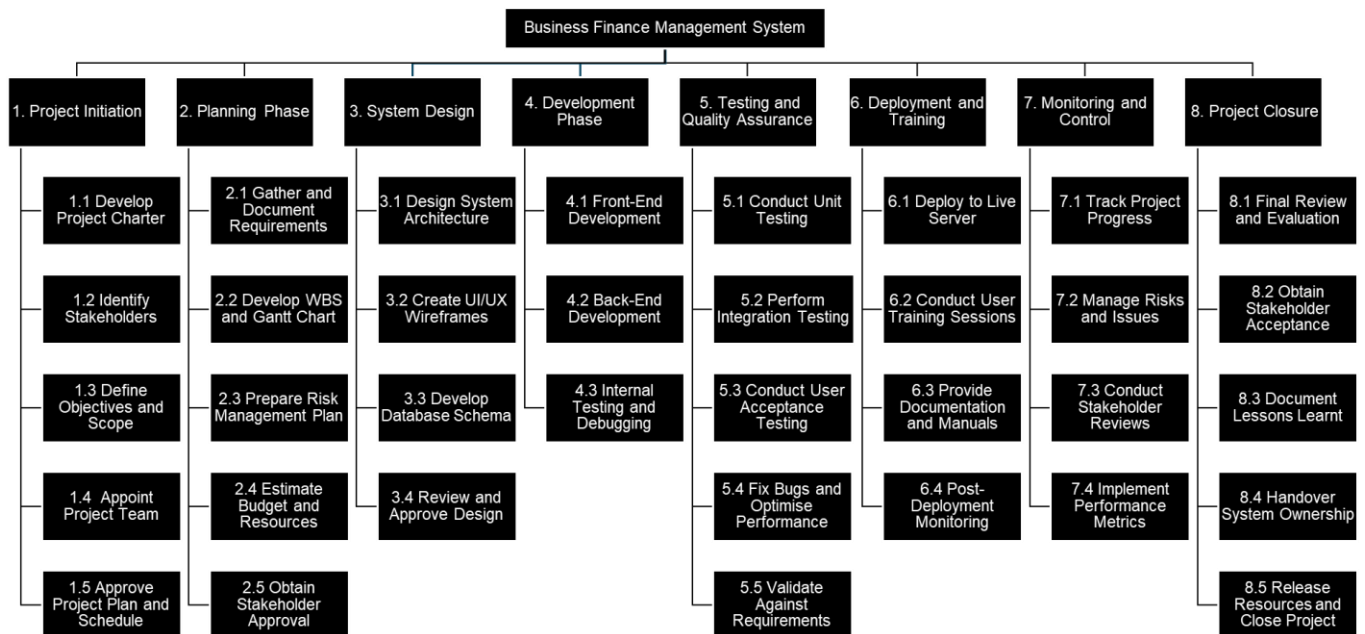


Figure 6 Work Breakdown Structure

Gantt Chart:

Task	March	April	May	June	July	August	September	October
Project Planning Initiation	■							
Planning Phase		■						
System Design			■					
Development Phase				■	■	■		
Testing and Quality Assurance						■		
Deployment and Training							■	
Monitoring and Control							■	
Project Closure								■

Figure 7 Gantt Chart

Agile Iteration Plan:

Methodology: Scrum-based Agile

Sprint Duration: 2 Weeks

Total Sprints: 8

Sprint 1: User Authentication & Core Setup

Goal: Implement secure user registration, login, and role-based access control (Admin, Staff).

User Stories:

- As a user, I can register an account to access the system.
- As an admin, I can assign roles to users to control access.
- As any user, I can securely log in to protect my data.
- Demo: Working login/register system with a basic admin dashboard.

Sprint 2: Income & Expense Management

Goal: Build functionality for recording and categorizing financial transactions.

User Stories:

- As a staff member, I can record income transactions to track revenue.
- As a staff member, I can record and categorize expenses to monitor spending.
- As a user, I can view a list of my transactions to review financial activity.
- Demo: Transaction entry forms and a basic transaction ledger view.

Sprint 3: Budgeting Module

Goal: Allow users to create budgets and track spending against them.

User Stories:

- As a user, I can create a monthly budget for different categories.
- As a user, I can see actual spending vs. budget in a dashboard.
- The system alerts me if I am nearing my budget limit.
- Demo: Budget creation interface and a simple dashboard with budget vs. actuals.

Sprint 4: Invoicing & Billing

Goal: Enable users to create, send, and track invoices.

User Stories:

- As a user, I can generate a professional invoice for a client.
- As a user, I can mark an invoice as paid to update my records.
- As a user, I can view a list of all sent invoices and their status.
- Demo: Invoice creation wizard and an invoice management list.

Sprint 5: Payroll Foundation

Goal: Develop core payroll setup and calculation engine.

User Stories:

- As an admin, I can add employees to the payroll system.
- The system automatically calculates gross pay based on hours/salary.
- The system calculates basic tax withholdings.
- Demo: Employee management screen and a preview of payroll calculations.

Sprint 6: Reporting & Dashboards

Goal: Implement financial reports and the main interactive dashboard.

User Stories:

- As a user, I can view a dashboard showing income, expenses, and cash flow.
- As an admin, I can generate a Profit & Loss statement.
- As a user, I can filter reports by date range.
- Demo: Interactive dashboard with charts and key financial metrics.

Sprint 7: Integration, Security & UAT Preparation

Goal: Integrate all modules, perform end-to-end testing, and harden security for UAT.

Key Tasks:

- Integrate payroll with reporting and general ledger.
- Conduct security penetration testing and vulnerability scans.
- Perform performance testing under load.
- Prepare UAT environment and test scripts.
- Demo: Fully integrated system ready for end-user testing.

Sprint 8: User Acceptance Testing & Bug Fixing

Goal: Validate the system with real users and fix critical issues before launch.

Key Tasks:

- Facilitate UAT sessions with selected SMEs.
- Collect and prioritize bug reports and feedback.
- Fix all critical and high-priority bugs.
- Finalize deployment and rollback plans.
- Outcome: UAT sign-off from stakeholders and a shippable product.

Post-Sprint: Deployment & Launch

Activities:

- Final deployment to the live production server.
- Data migration from legacy systems (if any).
- Conduct end-user training sessions.
- Go-live and begin post-launch support.

Milestone Schedule with Acceptance Criteria:

Milestone	Completion Date	Stakeholder Judge	Acceptance Criteria
Planning & Requirement Analysis	30/04/2025	John Smith (Project Manager)	Business needs completely decomposed.
		Sarah Johnson (Business Analyst)	High-level documentation of all the features needed.

		Michael Brown (CEO)	Project priorities and scope aligned with stakeholders.
Design & Development	31/06/2025	Emily Davis (Lead Designer)	Stakeholder-signed-off wireframes and UI/UX.
		Daniel Lee (Backend Developer)	Backend architecture and database schema defined.
Testing & Security Implementation	31/08/2025	Kevin White (QA Lead)	Entire system testing (unit, integration, and user test) done.
		Michael Brown (CEO)	Bug fixing and performance tuning.
Deployment & Training	30/09/2025	Olivia Green (Training Coordinator)	System successfully deployed and accessible by end users.
		Michael Brown (CEO)	Final review and feedback incorporated.

Figure 8 Milestone Schedule

Project Risks:

Project Risks	Risk Owner	Contingency Plans
1. Delays in development due to technical challenges.	Daniel Lee (Backend Developer)	Do frequent code review and sprint retrospectives.
2. Budget overruns due to scope creep.	John Smith (Project Manager)	Set firm scope boundaries and feature prioritization.
3. Integration issues with third party systems.	Emily Davis (Lead Designer)	Plan for a dedicated testing phase for integrations.

4. Low user adoption due to resistance to change.	Sarah Johnson (Business Analyst)	Conduct training sessions, provide user guides, and gather feedback during pilot testing.
5. Data security and privacy breaches.	Michael Brown (IT Security Lead)	Implement strong encryption, role-based access controls, and regular security audits.

Figure 9 Project Risks

Resources Plan

RACI Chart:

Deliverable	John Smith (Project Manager)	Michael Brown (CEO)	Sarah Johnson (Business Analyst)	Emily Davis (Lead Designer)	Daniel Lee (Backend Developer)	Kevin White (QA Lead)
Project Charter	A	C	R	I	I	I
Requirements Documentation	A	C	R	C	C	I
System Architecture & UI/UX Design	I	C	C	R	C	I
Front-End Development	I	I	I	R	C	C
Back-End & Database Development	I	I	I	C	R	C
Unit & Integration Testing	I	I	I	I	R	A

User Acceptance Testing (UAT)	A	C	R	C	C	R
User Training & Documentation	A	I	C	I	I	I
System Deployment	A	I	I	I	R	R
Project Closure & Handover	R	A	I	I	I	I
Budget & Schedule Control	A	C	I	I	I	I
Risk & Issue Management	A	C	C	C	C	C

Figure 10 RACI Chart

Operational Estimates:

Category	Amount	Description
Customer Support Services	R7 500	Helpdesk/ticketing system to support SMEs after implementation.
Hosting and Maintenance	R3000	Cloud servers, database administration, and server upkeep.
Marketing & Onboarding Support	R10 000	Business migration support, training, and implementation assistance.
Total	R20 500	

Figure 11 Operational Estimates

People Estimates:

Category	Amount	Description
Project Manager	R120 000	Oversees project delivery, manages timelines, and coordinates team.
Backend Developers (2)	R190 000	Develop server-side logic, manage databases, and ensure scalability.
QA Specialist	R75 000	Tests system performance, compliance, and functionality.
Total	R385 000	

*Figure 12 People Estimates***Equipment and Other:**

Category	Amount	Description
Development Workstations (3)	R30 000	Coding and testing hardware for developers.
Software Licenses	R5 000	Development tools, version control, database management, and project management software.
Office/Collaboration Tools	R6 000	Communication and productivity tools.
Total	R41 000	

Figure 13 Equipment and Other Estimates

Stakeholder Analysis & Engagement

Stakeholder List:

	Stakeholder	Role	Interest in Project	Influence
Primary:				
	John Smith	Project Manager	Manages the project's progress, ensuring milestones are met and maintaining budget control.	High
	Michael Brown	CEO	Ensures the system aligns with business goals and contributes to improved financial management.	High
	Daniel Lee	Backend Developer	Focuses on building the backend architecture and ensuring the functionality meets business requirements.	Medium
	Emily Davis	Lead Designer	Designs an intuitive UI/UX that meets user expectations and provides a smooth experience.	Medium
Others:				
	Sarah Johnson	Business Analyst	Ensures the system captures all business requirements and aligns with financial management goals.	High

	Olivia Green	Training Coordinator	Ensures that end-users, admins, and employees are trained to use the system effectively.	Medium
	Kevin White	QA Lead	Ensures the system is free of defects, performs well, and meets all quality standards.	Medium

Figure 14 Stakeholder List

Stakeholder Engagement Plan:

Stakeholder	Communication Method	Frequency	Purpose
John Smith (Project Manager)	Daily stand-ups, project dashboards	Daily	Monitor progress, resolve issues
Michael Brown (CEO)	Progress reports, milestone reviews	Monthly	Review overall performance, ensure alignment with business goals
Daniel Lee (Backend Developer)	Developer sync meetings	Twice weekly	Track development progress, address technical blockers
Emily Davis (Lead Designer)	Design review sessions, prototype testing	Weekly	Ensure UI/UX meets stakeholder expectations
Sarah Johnson (Business Analyst)	Requirements review meetings	Weekly	Validate business requirements and change requests
Olivia Green (Training Coordinator)	Coordination meetings, email updates	Bi-weekly	Prepare user training materials and onboarding plans
Kevin White (QA Lead)	QA review meetings, defect reports	Weekly	Monitor quality control activities and testing progress

Figure 15 Stakeholder Engagement

Stakeholder Engagement Tools and Channels

- Email & Project Management Tools: For formal communication, document sharing, and tracking updates.
- Microsoft Teams / Zoom: For virtual meetings and live demonstrations.
- Project Dashboard: To track task progress, risks, and milestones.
- Reports & Presentations: For status updates to management and key decision-makers.
- Workshops & Training Sessions: To promote active engagement and system understanding.

Team Operating Principles:

1. **Effective Meetings:** All meetings should be well-planned, start and end on time, and have a clear agenda to ensure productivity and respect for everyone's time.
2. **Active Participation:** Team members are encouraged to contribute ideas, participate in discussions, and provide constructive feedback. Differences in perspective should be respected and used to strengthen project outcomes.
3. **Collaboration:** Work should be broken down into manageable components with assigned ownership. Regular progress checks and open communication should ensure accountability and alignment with project goals.
4. **Positive and Respectful Culture:** Team members should maintain a supportive and professional environment, embrace constructive criticism and foster positive relationships to enhance teamwork and project success.
5. **Continuous Improvement:** The team should regularly reflect on processes, identify areas for improvement, and adopt best practices to enhance efficiency, quality, and overall project outcomes.

Lessons Learned:

Scope creep occurred in Business Finance Management System development when new features like new reporting calculations were introduced after final requirements were finalised. The project timeline was stretched, and it went over budget. In future projects, any change in scope must be thoroughly analysed and documented by all stakeholders so that any changes are in line with the original project goals are agreed upon.

Signatures and Commitment:

Sponsor	Department / Organization	Signature
Jessica Thompson	Finance Department	
Project Manager	Department / Organization	Signature
John Smith	Project Management Office	
Core Team Members	Department/ Organization	Signature
Michael Brown	Executive Leadership	
Daniel Lee	Development Team	
Emily Davis	Design Team	

Figure 16 Signatures and Commitment

Risk Management Plan

Business Finance Management Project

Date 10 October 2025

Risk Register:

Category of Risk	Specific Risk	Potential Cause(s)	Potential Response(s)	Owner	Trigger	Response Strategy
Technical	Development delays	Integration errors or complex code	More code reviews and team checks	Daniel Lee (Backend Developer)	Missed deadlines	Mitigation
Financial	Budget overruns	Scope creep or underestimated resource costs	Enforce scope control and approvals	John Smith (Project Manager)	Budget nearly exceeded	Prevention
Integration	Third-party issues	API incompatibility or lack of proper documentation	Early testing and documentation	Emily Davis (Lead Designer)	Integration errors	Mitigation
Staffing	Key staff unavailable	Illness, resignation, or personal emergencies	Crosstrain staff and back up roles	John Smith (Project Manager)	Missed deliverables	Contingency
Operational	Low user adoption	Insufficient training or resistance to change	Training and post-launch support	Olivia Green (Training Coordinator)	Low usage rates	Mitigation

Figure 17 Risk Register

RAID Log:

Category	Description	Owner	Impact	Status	Action Plan
Risk	Delays in backend development due to integration challenges.	Daniel Lee (Backend Developer)	High	Active	Conduct frequent integration tests; allocate extra buffer time in the schedule.

	Possible data security breaches during user authentication.	Kevin White (QA Lead)	High	Active	Implement strong encryption, SSL certificates, and regular vulnerability checks.
	Incomplete user adoption after system launch.	Olivia Green (Training Coordinator)	Medium	Monitoring	Provide proper training sessions, guides, and ongoing user support.
Assumption	Users will have reliable internet access to use the web system.	John Smith (Project Manager)	Medium	Validated	Confirmed through user survey during planning stage.
	The project budget will remain within the approved financial limits.	Michael Brown (CEO)	High	Valid	Ongoing budget monitoring by project manager.
Issue	Minor bugs discovered during system testing caused deployment delays.	Kevin White (QA Lead)	Medium	Resolved	Extended testing phase by one week to address defects.
	Late delivery of design assets from the UI/UX team.	Emily Davis (Lead Designer)	Low	Closed	Improved communication and early submission deadlines.
Dependency	Successful connection to the database depends on proper hosting setup.	Daniel Lee (Backend Developer)	High	Completed	Hosting server configured and tested successfully.
	Payroll and invoicing modules rely on accurate financial data from users.	Sarah Johnson (Business Analyst)	Medium	Active	Include data validation and import templates to reduce errors.

Figure 18 RAID Log

Quality Management Plan

Business Finance Management Project

Date 10 October 2025

Project Overall Quality Objectives

The Business Finance Management System (BFMS) project aims to deliver a secure, reliable, and user-friendly financial management platform for SMEs. The specific quality objectives are:

- Ensure the BFMS meets all functional and non-functional requirements with $\leq 2\%$ defect leakage post-deployment.
- Deliver a highly usable system with 80% user adoption within 6 months of go-live.
- Maintain compliance with financial regulations and data security standards.
- Guarantee system availability of at least 99% uptime.
- Provide comprehensive documentation and training resources to support smooth adoption.

Quality Standards

The BFMS project will be judged against the following standards:

- International Standards: ISO 9001 for quality management processes, ISO 27001 for information security, and IFRS compliance for financial reporting.
- Coding Standards: Adherence to clean coding principles, modular design, and consistent version control practices.
- Usability Standards: Interfaces must comply with basic accessibility guidelines and support responsive design.
- Performance Standards: The system must process transactions within 2 seconds on average and handle peak usage without downtime.

Quality Tools and Methodologies

The project will apply recognised quality management tools and practices:

- Total Quality Management (TQM): Continuous monitoring and improvement of processes, with a focus on customer satisfaction and defect prevention.
- Six Sigma: Defining measurable quality metrics, analysing variance, and applying DMAIC (Define, Measure, Analyse, Improve, Control) for defect reduction.

- ISO Standards: Leveraging ISO 9001 and ISO 27001 as benchmarks for quality and security.
- Quality Control Tools: Defect tracking systems, automated testing tools, and peer code reviews will be used to ensure ongoing compliance.

Integration with Project Processes

Quality management will be embedded throughout the project lifecycle:

- Requirements Phase: A traceability matrix will ensure all client requirements are captured and validated.
- Development Phase: Continuous integration, code reviews, and automated unit testing will maintain coding and performance standards.
- Testing Phase: Formal test plans, system testing, and user acceptance testing (UAT) will ensure all deliverables meet agreed criteria.
- Deployment Phase: Staging environments will be used for final quality checks before production release.
- Post-Deployment: Monitoring dashboards and SLA-driven support processes will track uptime, performance, and issue resolution.

Project Deliverables

Deliverable	Standard for Judging Each Deliverable from Customer Perspective
Functional BFMS System	System must perform all core functions (transaction recording, payroll, invoicing, reporting) without critical defects.
User Interface and Dashboard	Intuitive, responsive, and simple-to-use; users can complete tasks with minimal training.
Documentation (User Manuals, Technical Guides)	Clear, concise, and comprehensive, supporting both technical staff and end-users.
Training Materials & Onboarding Support	Easy to follow; ensures 80% of users can operate the system effectively within 2 weeks of training.
Maintenance & Support Plan	Reliable support channels, clear SLAs, and regular updates.

Figure 19 Project Deliverables

Work Process

Process	Standard for Judging Each Process
Requirements Gathering	Requirements must be validated and signed off by stakeholders before development begins.
Development	Code should follow clean coding practices, undergo peer reviews, and be version controlled.
Testing	Each module must pass unit, integration, and user acceptance testing with $\leq 2\%$ defect leakage.
Deployment	Deployment must be documented, tested in staging first, and cause zero downtime for clients.
Maintenance	Issues must be logged, prioritised, and resolved within agreed SLAs.

Figure 20 Work Process

Quality Control Activities

- Conduct unit testing and integration testing during development.
- Regular code reviews to ensure adherence to coding standards.
- System testing to validate performance, security, and compliance.
- User Acceptance Testing (UAT) with SMEs before final deployment.
- Defect tracking and resolution through a structured issue management tool.

Transition & Closure Plan

Business Finance Management Project

Date 10 October 2025

Current Project Situation

The Business Finance Management System (BFMS) project has been successfully completed. All core functionalities including budgeting, reporting, payroll, and invoicing have been fully developed, tested, and deployed. The system is now operational, with users actively engaging in day-to-day financial management through the new platform.

Schedule of Remaining Work

All planned work has been completed, including documentation, user training, system optimisation, and deployment. A post-implementation review has been scheduled to evaluate system performance, user adoption, and alignment with business goals. Any identified improvements will be logged for potential future system updates.

Risks Remaining

- User adaptation risk: Some users may still require additional support to become fully confident using the system.
- Post-launch maintenance: Minor technical issues may arise during extended use, requiring timely developer intervention.
- Data accuracy monitoring: Ongoing checks are necessary to ensure all migrated financial data remains consistent and up to date.

Recommendations to Users of Project Deliverables

Users should continue referring to the System Documentation and User Manual for operational guidance. Refresher training sessions and ongoing technical support are available through the helpdesk to ensure smooth adoption. Users are encouraged to provide feedback for any challenges or feature requests to guide continuous improvement.

Instructions for Measuring Benefits

The system's success should be evaluated through:

- Reduction in manual data entry and processing time

- Improvement in financial accuracy and reporting speed
- Higher compliance rates with financial regulations
- Increased user satisfaction and operational efficiency

Quarterly performance reviews will assess these benefits to ensure the system continues to deliver value to the organisation.

Transition Strategy

The transition to operational use was completed through a structured rollout that included:

1. Deployment: The system was successfully installed and configured on live servers.
2. Training & Handover: Users and administrators were trained and given full operational control.
3. Support Period: A one-month post-launch support phase ensured system stability and addressed early feedback.

Success Definition

The project is considered successful as it achieved all planned deliverables, met quality standards, and received positive user feedback. The system has improved financial reporting accuracy, reduced administrative workload, and streamlined operations across departments.

Lessons Learned

Throughout the development and implementation of the Business Finance Management System (BFMS), several valuable lessons were identified that can guide future projects of a similar nature. One of the most significant lessons was the importance of maintaining clear scope boundaries. Early in the project, scope adjustments and feature requests—such as the inclusion of additional reporting options—led to time extensions and budget pressures. Establishing a strict scope management process and ensuring all change requests go through a formal evaluation helped prevent further overruns.

Another major insight was the need for early and consistent stakeholder involvement. Engaging key users, including financial officers and administrators, from the initial planning stages ensured that the system met actual business needs rather than assumptions. Regular feedback loops through sprint reviews and prototype demonstrations allowed for faster adjustments and stronger alignment with user expectations.

Finally, the project emphasised the importance of comprehensive documentation and training. Providing detailed user manuals, admin guides, and live training sessions ensured that end-users could adapt quickly, reducing post-deployment support demands. These lessons will serve as key reference points for improving planning, execution, and delivery in future digital transformation initiatives.

Feedback

Post-deployment feedback from project sponsors, developers, and end-users indicated high satisfaction with the system's ease of use, automation features, and real-time dashboards. Users reported reduced manual work in payroll, invoicing, and reporting, while management appreciated improved accuracy and compliance with financial standards.

Some users suggested adding advanced analytics and custom reporting in future updates, as well as a mobile-friendly interface for easier access. Others recommended periodic refresher training for new staff to maintain efficient use. Overall, feedback confirmed that the BFMS achieved its key goals and provided a strong foundation for continued improvement and scalability.

Lessons Learnt & Reflection

The development of the Business Finance Management System (BFMS) provided significant learning opportunities and reinforced the practical application of key Project Management Body of Knowledge (PMBOK) principles. One of the most valuable lessons was the importance of effective scope and time management. Early feature additions highlighted how easily scope creep can disrupt project flow, reinforcing the need for a well-defined Scope Baseline and controlled Change Management Process as outlined in PMBOK's Integration and Scope Management knowledge areas.

Another key learning outcome was the significance of stakeholder engagement and communication. Regular meetings, progress updates, and feedback sessions ensured that stakeholder needs were met and that expectations remained aligned with project objectives. This experience strengthened my understanding of the Stakeholder and Communications Management principles, particularly in maintaining transparency and fostering collaboration throughout the project lifecycle.

The project also deepened my appreciation for Risk and Quality Management. By identifying potential risks early such as technical delays and integration challenges and implementing

mitigation strategies, the project-maintained stability and avoided major disruptions. Continuous testing and quality reviews ensured the system met both user expectations and performance standards, reflecting PMBOK's emphasis on proactive quality assurance.

Finally, I learned the value of team coordination and leadership in achieving project success. Balancing technical, design, and managerial perspectives required flexibility, empathy, and clear decision-making. The experience not only strengthened my technical project management skills but also enhanced my ability to lead cross-functional teams effectively.

Overall, this project was a practical demonstration of PMBOK principles in action, highlighting that successful project delivery relies as much on structured processes as on communication, adaptability, and continuous learning.

Cape Peninsula University of Technology

Department of Information Technology

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Due Date: _____ 10 October 2025 _____

Lecturer: _____ Ayodeji Olanrewaju _____

I am submitting the assignment for:

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I/We declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged; (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration; and (iii) if applicable, the submitted soft copy of the assignment is identical to the hard copy to be submitted.

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